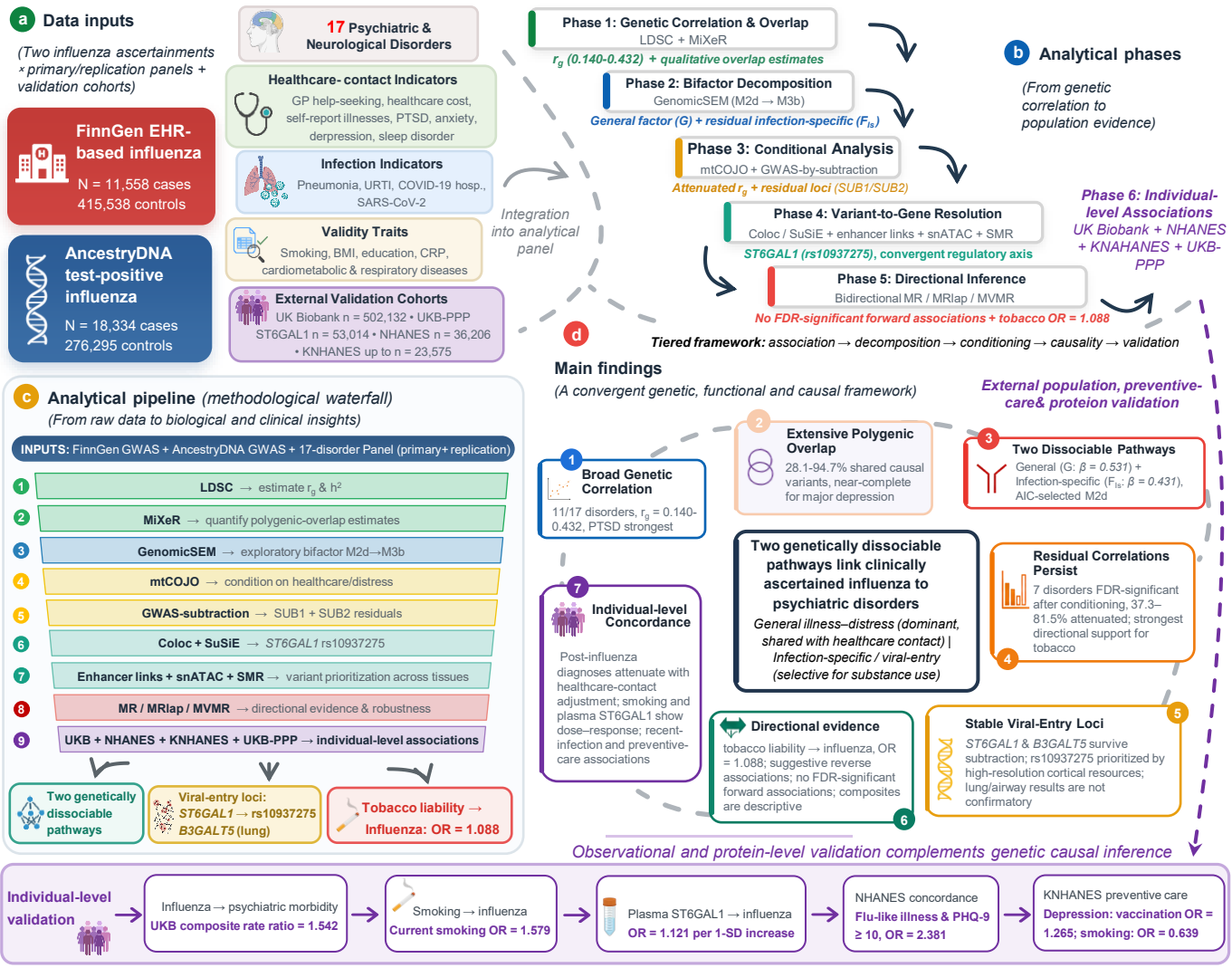

a Data inputs
(Two influenza ascertainments × primary/replication panels + validation cohorts)
FinnGen EHR-based influenza
N = 11,558 cases
415,538 controls
AncestryDNA test-positive influenza
N = 18,334 cases
276,295 controls
17 Psychiatric & Neurological Disorders
Healthcare- contact Indicators
GP help-seeking, healthcare cost, self-report illnesses, PTSD, anxiety, derpression, sleep disorder
Infection Indicators
Pneumonia, URTI, COVID-19 hosp., SARS-CoV-2
Validity Traits
Smoking, BMI, education, CRP, cardiometabolic & respiratory diseases
External Validation Cohorts
UK Biobank n = 502,132 • UKB-PPP ST6GAL1 n = 53,014 • NHANES n = 36,206 • KNHANES up to n = 23,575
Integration into analytical panel
b Analytical phases
(From genetic correlation to population evidence)
Phase 1: Genetic Correlation & Overlap
LDSC + MiXeR
rg (0.140-0.432) + qualitative overlap estimates
Phase 2: Bifactor Decomposition
GenomicSEM (M2d → M3b)
General factor (G) + residual infection-specific (Fls)
Phase 3: Conditional Analysis
mtCOJO + GWAS-by-subtraction
Attenuated rg + residual loci (SUB1/SUB2)
Phase 4: Variant-to-Gene Resolution
Coloc / SuSiE + enhancer links + snATAC + SMR
ST6GAL1 (rs10937275), convergent regulatory axis
Phase 5: Directional Inference
Bidirectional MR / MRlap / MVMR
No FDR-significant forward associations + tobacco OR = 1.088
Phase 6: Individual-level Associations
UK Biobank + NHANES + KNAHANES + UKB-PPP
Tiered framework: association → decomposition → conditioning → causality → validation
c Analytical pipeline (methodological waterfall)
(From raw data to biological and clinical insights)
INPUTS: FinnGen GWAS + AncestryDNA GWAS + 17-disorder Panel (primary+ replication)
1 LDSC → estimate rg & h²
2 MiXeR → quantify polygenic-overlap estimates
3 GenomicSEM → exploratory bifactor M2d→M3b
4 mtCOJO → condition on healthcare/distress
5 GWAS-subtraction → SUB1 + SUB2 residuals
6 Coloc + SuSiE → ST6GAL1 rs10937275
7 Enhancer links + snATAC + SMR → variant prioritization across tissues
8 MR / MRlap / MVMR → directional evidence & robustness
9 UKB + NHANES + KNHANES + UKB-PPP → individual-level associations
Two genetically dissociable pathways
Viral-entry loci: ST6GAL1 → rs10937275 B3GALT5 (lung)
Tobacco liability → Influenza: OR = 1.088
d Main findings
(A convergent genetic, functional and causal framework)
External population, preventive-care& proteion validation
1 Broad Genetic Correlation
11/17 disorders, rg = 0.140-0.432, PTSD strongest
2 Extensive Polygenic Overlap
28.1-94.7% shared causal variants, near-complete for major depression
3 Two Dissociable Pathways
General (G: β = 0.531) + Infection-specific (Fls: β = 0.431), AIC-selected M2d
Two genetically dissociable pathways link clinically ascertained influenza to psychiatric disorders
General illness–distress (dominant, shared with healthcare contact) | Infection-specific / viral-entry (selective for substance use)
4 Residual Correlations Persist
7 disorders FDR-significant after conditioning, 37.3–81.5% attenuated; strongest directional support for tobacco
5 Stable Viral-Entry Loci
ST6GAL1 & B3GALT5 survive subtraction; rs10937275 prioritized by high-resolution cortical resources; lung/airway results are not confirmatory
6 Directional evidence
tobacco liability → influenza, OR = 1.088; suggestive reverse associations; no FDR-significant forward associations; composites are descriptive
7 Individual-level Concordance
Post-influenza diagnoses attenuate with healthcare-contact adjustment; smoking and plasma ST6GAL1 show dose–response; recent-infection and preventive-care associations
Observational and protein-level validation complements genetic causal inference
Individual-level validation
Influenza → psychiatric morbidity
UKB composite rate ratio = 1.542
Smoking → influenza
Current smoking OR = 1.579
Plasma ST6GAL1 → influenza
OR = 1.121 per 1-SD increase
NHANES concordance
Flu-like illness & PHQ-9 ≥ 10, OR = 2.381
KNHANES preventive care
Depression: vaccination OR = 1.265; smoking: OR = 0.639